

CLOUD HOSTING BUDGET ESTIMATE

Agentic BuilderX Production Platform

A non-binding monthly operating estimate for an AWS Mumbai deployment.

REGION AWS Mumbai	CAPACITY 25 concurrent builds	VOLUME 5,000 builds/month
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EXPECTED MONTHLY BUDGET \$3,400-\$4,400 USD, including infrastructure and expected OpenAI model usage.

This estimate is designed for a multi-AZ launch serving approximately 100-300 active users. **It is not a binding quotation or invoice.**

Executive recommendation

Deploy BuilderX as a managed control plane with isolated, queue-driven build workers. The existing local Docker Compose topology is suitable for development, but it should not be exposed directly as a public multi-user service.

- Serve the BuilderX frontend and static previews through CloudFront and S3.
- Run the API as stateless ECS services across at least two availability zones.
- Queue builds in SQS and execute each build in an isolated, time-limited worker task.
- Store users, projects, builds, permissions, decisions, and snapshots in managed databases.
- Store generated files, media, exports, and workspace archives in S3.

Estimated monthly budget

Cost category	Expected range	What it covers
API and control plane	\$90-\$180	Multi-AZ ECS services for the BuilderX API and orchestration control plane.
Build and preview workers	\$150-\$400	Approximately 5,000 isolated builds, queued overflow, and time-limited previews.
Aurora PostgreSQL	\$180-\$350	Project ownership, workflow state, build records, permissions, and snapshots.
Redis and queues	\$60-\$140	Live workflow state, event distribution, coordination, and SQS requests.
Network and load balancing	\$100-\$250	ALB, private connectivity, routing, and typical transfer.
Storage and delivery	\$50-\$180	S3, CloudFront, ECR, build artifacts, exports, and backups.
Security and operations	\$100-\$300	WAF, CloudWatch, audit logging, alarms, and backup operations.
Neo4j AuraDB	About \$292	Business Critical minimum for the production agent relationship graph.
Infrastructure subtotal	\$970-\$1,940	Excludes model usage, engineering labour, taxes, and premium support.

AI model usage

Model usage is the largest variable. The estimates below use published GPT-5 pricing of \$1.25 per million input tokens and \$10 per million output tokens. Actual coding-agent usage depends on task complexity, retries, reviewers, context size, and caching.

Scenario	Tokens per build	Approx. monthly AI cost
Efficient	75k input + 15k output	\$1,220
Expected	150k input + 30k output	\$2,438
Heavy	300k input + 60k output	\$4,875

COST-CONTROL OPPORTUNITY Use smaller models for classification, routine edits, and low-risk review. Reserve GPT-5-class execution for complex or high-risk work. A hybrid route could reduce expected AI usage to roughly \$700-\$1,500 per month.

Required production work

- 1 Remove Docker-socket access and replace host-created preview containers with isolated cloud tasks.
- 2 Replace local JSON, JSONL, logs, workspaces, and project registries with managed persistence.
- 3 Cryptographically verify Google identity tokens and enforce ownership on every project resource.
- 4 Convert synchronous generation into durable queued jobs with status, cancellation, retries, and idempotency.
- 5 Build production frontend assets; do not operate public Vite development servers.
- 6 Add quotas, rate limits, audit logs, malware scanning, worker network controls, monitoring, and tested backups.

Architecture at a glance

Layer	Recommended service	Responsibility
Edge	Route 53, ACM, CloudFront, WAF	DNS, TLS, caching, and perimeter protection.
Application	ALB + ECS Fargate	Authenticated API and orchestration control plane.
Build execution	SQS + isolated ECS tasks	Sandboxed agent execution with bounded resources.
Data	Aurora PostgreSQL + Redis	Durable metadata and short-lived coordination state.
Knowledge	Neo4j AuraDB + vector store	Agent relationships and reusable memory.
Artifacts	S3 + CloudFront	Generated files, exports, media, snapshots, and previews.

Assumptions and exclusions

- AWS Mumbai, multi-AZ operation, 100-300 active users, and 25 concurrent builds.
- Approximately 5,000 build instructions per month and a ten-minute average worker duration.
- Static previews are preferred; dynamic previews expire after 30 minutes of inactivity.
- Amounts exclude implementation labour, GST or other taxes, premium support, unusually high egress, and third-party observability subscriptions.
- Provider rates and exchange rates may change; confirm the final design in the AWS Pricing Calculator before purchase.

VALIDITY Budgetary estimate valid for planning purposes for 30 days from 5 July 2026. Final costs are usage-based and charged directly by service providers.

Reference pricing

Amazon Lightsail instance bundles:

<https://docs.aws.amazon.com/lightsail/latest/userguide/amazon-lightsail-bundles.html>

AWS Fargate pricing: <https://aws.amazon.com/fargate/pricing/>

AWS Elastic Load Balancing pricing: <https://aws.amazon.com/elasticloadbalancing/pricing/>

OpenAI GPT-5 developer pricing: <https://openai.com/index/introducing-gpt-5-for-developers/>

Neo4j AuraDB pricing: <https://neo4j.com/pricing/>

Budget decision

For a private beta, a single hardened virtual machine can reduce initial infrastructure cost significantly. For the selected 25-concurrent-build production target, the managed architecture above is the safer baseline because it separates users, build workers, data, and credentials.